

MODULE 01

Introduction to SQL and Databases

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

Introduction to SQL and Databases

What is SQL?

SQL (Structured Query Language) is the standard language for relational databases. Powers MySQL, PostgreSQL, Oracle, and SQL Server. Used by developers, analysts, and data scientists worldwide.

Key Concepts

Database: organised collection of data
Table: data in rows and columns
Row: a single record
Column: an attribute/field
Primary Key: unique identifier per row
Query: request to retrieve or change data

Basic Example

```
SELECT first_name, last_name, gpa
FROM students
WHERE gpa > 3.5
ORDER BY gpa DESC
LIMIT 10;
```

SQL vs NoSQL

SQL: structured schema, ACID, complex queries (MySQL, PostgreSQL)
NoSQL: flexible schema, horizontal scale (MongoDB, Redis, Cassandra)
SQL: banking, ERP, e-commerce
NoSQL: social media, IoT, real-time

Why Learn SQL?

SQL is the most in-demand data skill globally.
Powers every major database system.
Essential for backend dev, data analysis, and data engineering.